

Grup	Nom de la variable	Descripció
	FL_targetTorque	Target torque motor Front Left [Nm]
	FL_targetVelocity	Target Angular Velocity motor [rad/s]
	FL_actualVelocity	Actual Angular Velocity motor [rad/s]
	FL_actualTorque	Actual Torque motor [Nm]
	FL_actualSCurrent	Actual S current [A] (mòdul)
	FL_actualDCurrent	Actual D current [A]
	FL_actualQCurrent	Actual Q current [A]
	FL_actualSVoltage	Actual S voltage [V]
	FL_actualDVoltage	Actual D voltage [V]
	FL_actualQVoltage	Actual Q voltage [V]
	FL_actualPower	Actual Power Filtered {Ia*Va + Ib*Vb + Ia*Vc} [W] (potencia elèctrica a l'inverter)
	FL_resolverSignal	Senyal del resolver
	FL_motorTemperature	Temperatura motor [°C]
	FL_inverterTemperature	Temperatura power module [°C]
	FL_StatusWord	Inverter Supervision status 1 (0x2900 / 0x04)
	FL_StatusWord2	Inverter Supervision status 2 (0x2900 / 0x06)
	FL_FWActive	Field Weakening Active
	FL_TargetReached	Target torque reached
FL motor data	FL_OverloadActive	Overload active
	FR_targetTorque	Same as FL
	FR_targetVelocity	
	FR_actualVelocity	
	FR_actualTorque	
	FR_actualSCurrent	
	FR_actualDCurrent	
	FR_actualQCurrent	
	FR_actualSVoltage	
	FR_actualDVoltage	
	FR_actualQVoltage	
	FR_actualPower	
	FR_resolverSignal	
	FR_motorTemperature	
	FR_inverterTemperature	
	FR_StatusWord	
	FR_StatusWord2	
	FR_FWActive	
	FR_TargetReached	
FR motor data	FR_OverloadActive	
	FL_IxTLoad	
	FR_IxTLoad	
	RL_targetTorque	Same as FL
	RL_targetVelocity	
	RL_actualVelocity	
	RL_actualTorque	
	RL_actualSCurrent	
	RL_actualDCurrent	
	RL_actualQCurrent	
	RL_actualSVoltage	
	RL_actualDVoltage	
	RL_actualQVoltage	
	RL_actualPower	
	RL_resolverSignal	
	RL_motorTemperature	
	RL_inverterTemperature	
	RL_StatusWord	
	RL_StatusWord2	
	RL_FWActive	
	RL_TargetReached	
RL motor data	RL_OverloadActive	
	RR_targetTorque	Same as FL
	RR_targetVelocity	
	RR_actualVelocity	

RR motor data	RR_actualTorque	
	RR_actualSCurrent	
	RR_actualDCurrent	
	RR_actualQCurrent	
	RR_actualSVoltage	
	RR_actualDVoltage	
	RR_actualQVoltage	
	RR_actualPower	
	RR_resolverSignal	
	RR_motorTemperature	
	RR_inverterTemperature	
	RR_StatusWord	
	RR_StatusWord2	
	RR_FWActive	
	RR_TargetReached	
	RR_OverloadActive	
	RL_IxTLoad	
	RR_IxTLoad	
Control Limitation Info	StatusLimitation	Status limitation Control Systems
Raw sensor data	Throttle	Throttle encoder value [0-100] (mean of two apps)
	Brake	Brake potentiometer value [0-100]
	Steering	Steering potentiometer value [rad] --> $*(180/\pi)*(16/26)$ per passar a recorregut potenciometre [mm]
Lap Info in DV (also in EV if SLAM is ON)	delta	Difference in lap times, from last lap to previous lap in DV (also in EV if SLAM is ON)
	dist_km	Distance the car has travelled in DV (also in EV if SLAM is ON)
	laps	Number of laps completed in DV (also in EV if SLAM is ON)
	laptime	Lap time in DV (also in EV if SLAM is ON)
Vectornag INS info INS: module that combines IMU and GPS measurements with a Kalman Filter	globalDelta	Difference in lap times, from last lap to best lap in DV (also in EV if SLAM is ON)
	insStatus	Vectornav INS Status
	numSats	Vectornav number of Satellite connected
	VN_latitude	Vectornav Latitude [m/s]
	VN_longitude	Longitud [m/s]
	VN_altitude	Altitude [m/s]
	VN_vx	Longitudinal Velocity (car reference frame) [m/s]
	VN_vy	Lateral Velocity [m/s]
	VN_vz	Vertical Velocity [m/s]
	RuleBased_vx	*revisar
	RuleBased_vy	
	VN_mode	Vectornav mode
Observers output	MKF_vx	Kalman filter longitudinal velocity based on IMU and angular wheel velocity [m/s]
	Est_Vy_dyn	Lateral velocity kalman filter [m/s]
	Est_Vy_kin	Lateral velocity kalman filter [m/s]
Vectornav IMU data	VN_ax	Vectornav longitudinal acceleration (Car reference frame, Vectornav frame) [m/s²]
	VN_ay	Lateral acceleration [m/s²]
	VN_az	Vertical acceleration [m/s²]
	VN_gz	Angular velocity Z axis [rad/s] (Yaw rate)
	VN_gx	Angular velocity X axis [rad/s] (Roll rate/velocity)
	VN_gy	Angular velocity Y axis [rad/s] (Pitch rate/velocity)
	VN_ox	Orientation Quaternion x
	VN_oy	Orientation Quaternion y
	VN_oz	Orientation Quaternion z
	VN_ow	Orientation Quaternion w
	Est_vxCOG	Longitudinal Velocity. Final velocity used in the LLC (may be from VN, limovelo, or observers) [m/s]
	Est_vyCOG	Lateral Velocity. Final velocity used in the LLC (may be from VN, limovelo, or observers) [m/s]
	Est_VXFL	Vx transformed to FL wheel frame [m/s]
	Est_VXFR	Vx transformed to FR wheel frame [m/s]
	Est_VXRL	Vx transformed to RL wheel frame [m/s]
	Est_VXRR	Vx transformed to RR wheel frame [m/s]
	Est_VYFL	Vy transformed to FL wheel frame [m/s]
	Est_VYFR	Vx transformed to FR wheel frame [m/s]
	Est_VYRL	Vx transformed to RL wheel frame [m/s]
	Est_VYRR	Vx transformed to RR wheel frame [m/s]
	Est_SRFL	Slip ratio FL wheel [0 - 1]
	Est_SRFR	Slip ratio FR wheel [0 - 1]
	Est_SRRL	Slip ratio RL wheel [0 - 1]

Kinematic Calculations	Est_SRRR	Slip ratio RR wheel [0 - 1]
	Est_SAFI	Slip angle FL wheel [rad]
	Est_SAFR	Slip angle FR wheel [rad]
	Est_SARL	Slip angle RL wheel [rad]
	Est_SARR	Slip angle RR wheel [rad]
Dynamic Calculations	Est_FXFL	Longitudinal Force FL [N]
	Est_FXFR	Longitudinal Force FR [N]
	Est_FXRL	Longitudinal Force RL [N]
	Est_FXRR	Longitudinal Force RR [N]
	Est_FYFL	Lateral Force FL [N]
	Est_FYFR	Lateral Force FR [N]
	Est_FYRL	Lateral Force RL [N]
	Est_FYRR	Lateral Force RR [N]
	Est_FZFL	Vertical Force FL [N]
	Est_FZFR	Vertical Force FR [N]
SLAM data	limovelo_x	Position x axis [m] (the (0,0) is the starting point)
	limovelo_y	Position y axis [m] (the (0,0) is the starting point)
	limovelo_vx	Longitudinal Velocity [m/s]
	limovelo_vy	Lateral Velocity [m/s]
SOC estimation	SoC	
Master Information	Master_frontLeftTrq	Master FL torque [Nm]
	Master_frontRightTrq	Master FR torque [Nm]
	Master_rearLeftTrq	Master RL torque [Nm]
	Master_rearRightTrq	Master RR torque [Nm]
	Plausability	Plausability situation On or Off
Torque Vectoring Info	TV_FL_Trq	Output motor torque [Nm]
	TV_FR_Trq	Output motor torque [Nm]
	TV_RL_Trq	Output motor torque [Nm]
	TV_RR_Trq	Output motor torque [Nm]
	TV_desiredYawRate	Reference yaw rate [rad/s]
	TV_errorYawRate	Error yaw rate [rad/s]
	TV_feedForwardMz	Angular moment in Z axis from Feed Forward [Nm]
	TV_feedBackMz	Mz from Feed Back [Nm]
	TV_desiredMz	Reference Mz [Nm]
	TV_actualMz	Actual Mz [Nm]
	TV_errorMz	Error Mz [Nm]
	TV_desiredFx	Reference Longitudinal force [N]
	TV_actualFx	Actual Fx [N]
	TV_errorFx	Error Fx [N]
	TV_alpha1	Weight from QP, that balances Fx
	TV_alpha2	Weight from QP, that balances Mz
	TV_Enable	Active control
	TV_Reset	Reset ON or Off
	LLC_Command	Final command for LLC (process bpe and throttle signals)
Regenerative Control Info	RB_FL_Trq	Output motor torque [Nm]
	RB_FR_Trq	Output motor torque [Nm]
	RB_RL_Trq	Output motor torque [Nm]
	RB_RR_Trq	Output motor torque [Nm]
	RB_F_HydTrq	*revisar
	RB_R_HydTrq	*revisar
	RB_F_TotalTrq	*revisar
	RB_R_TotalTrq	*revisar
	RB_Enable	Enable control (active when control is ON)
	RB_intensityTarget	Reference Intensity Control [A]
	TC_FL_MaxAngVel	Max Angular Velocity Motor [rad/s]
	TC_FR_MaxAngVel	Max Angular Velocity Motor [rad/s]
	TC_RL_MaxAngVel	Max Angular Velocity Motor [rad/s]
	TC_RR_MaxAngVel	Max Angular Velocity Motor [rad/s]
	TC_FL_MaxTrq	Output Motor Torque [Nm]
	TC_FR_MaxTrq	Output Motor Torque [Nm]
	TC_RL_MaxTrq	Output Motor Torque [Nm]
	TC_RR_MaxTrq	Output Motor Torque [Nm]

Traction Control Info	TCenable	Traction Control Enable Mode
	TCenableTorque	Torque Mode enable
	TCenableVelocity	Velocity Mode enable
Power Control Info	PC_FL_Trq	Output Motor Torque [Nm]
	PC_FR_Trq	Output Motor Torque [Nm]
	PC_RL_Trq	Output Motor Torque [Nm]
	PC_RR_Trq	Output Motor Torque [Nm]
	PCenable	Enable control (active when control is ON)
Mode Parameters Info	Param_desiredMaximumVelocity	Maximum motor velocity in rad/s
	Param_desiredMaximumTorque	Maximum car torque in Nm
	Param_desiredMaximumRegenTorque	Maximum car regen torque in Nm
	Param_motorOn	Number of motors on
	TVenableMode	Enable Torque Vectoring
	TCenableMode	Enable Traction Control
	PCenableMode	Enable Power Control
	RBenableMode	Enable Regenerative Control
	TV_tunning	*revisar
	TV_Kp_factor	*revisar
Dashboard Car Mode	TV_Ki_factor	*revisar
AS commands	CarMode	Dashboard Car mode
	AS_Motor	Motor command [-1 , 1]
AS Debug Info	AS_Steering	Steering command [rad]
	AS_yaw_rate	Yaw rate reference [rad/s]
	LateralError	Actual lateral error
	exitflagsNLOP	
	timeNLOP	comput time mpc
	timePalanner	Comput time planner
	timeLongPID	Comput time long pid
	targetVelocity	Target velocity [m/s]
PU	CPU_temp	
	CPU_load	
	RAM_load	
	WIFI_temp	
Battery Status	Vbat	Battery Voltage [V]
	Current	Battery Current [A]
	Vmin	Min cell voltage [V]
	Vmax	Max cell voltage [V]
	Tmax	Max cell Temperature [°C]
	Tavg	Average cell Temperature [°C]
	VminCell_ID	ID of cell with min voltage
	VmaxCell_ID	ID of cell with max voltage
	TmaxCell_ID	ID of cell with max temperature
	Capacitor_Voltage	
	IMD_Error	
	BMS_Error	
	AIRPositive	
	FinalRelay	
	Precharge	
	CapacitorPrecharge	
	CapacitorShort	
	SDCRoot	
	MainHoopShutdownButtons	
	MotorRL	
	MotorRR	
	BSPD	
	Interlocks	
	IMD_AMS	
	RES	
	CockpitShutdownButton	
	AS	
	Inertia	
	MotorFL	
	MotorFR	
	BOTS	

Electronics Info	TSMS	
	ACCU_FINAL_IN	
	ASMS_AS	
	RES_AS	
	EBS_Supply	
Damper positions	DampFL	
	DampFR	
	DampRL	
	DampRR	
Last SDC error	LastError	
Pressures brake circuits	BSEFront	Brakes pressures [bar]
	BSErear	Brakes pressures [bar]
Can info	can0_errorFrameCount	
	can0_errorFrame_rate	
	can1_errorFrameCount	
	can1_errorFrame_rate	
	can2_errorFrameCount	
	can2_errorFrame_rate	
	can3_errorFrameCount	
	can3_errorFrame_rate	
More Electronics info	Fecu_Temp	
	Migjorn_MCU_Temp	
	Migjorn_PCB_Temp	
	ASB_Temp	
	RES_RADIO_QUALITY	